Image Reconstruction with Variational Networks: Application to Synchrotron Radiation Imaging

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***Abstract*—In computed tomography, image reconstructed from limited projection data is subject to strong noise and artifacts. Reducing the number of projection views in synchrotron radiation imaging has a great benefits in decreasing the computation cost for image formation. Moreover, it also prevents the damage of biological specimen caused by the x-ray radiation exposure. In this paper, a new procedure for image reconstruction from limited projection views with the assist of deep learning approach is proposed. A deep learning approach is employed to estimate an initial guess of the tomographic image followed by a fast row-action reconstruction to guarantee the restoration of any pattern missed in the first stage. Most of previous attempts that use deep learning in image reconstruction are implemented as a post-processing techniques applied on the reconstructed image. This may lead to anomalies in the final result due to the lack of consistency between the acquired projection data and the reconstructed image. The proposed procedure solves this problem by keeping the data consistency up to the final stage of image formation. This would reduce the possibility of losing image abnormal structures due to insufficient training in deep learning. Experimental results using synchrotron radiation data demonstrate the usefulness of the proposed framework.**

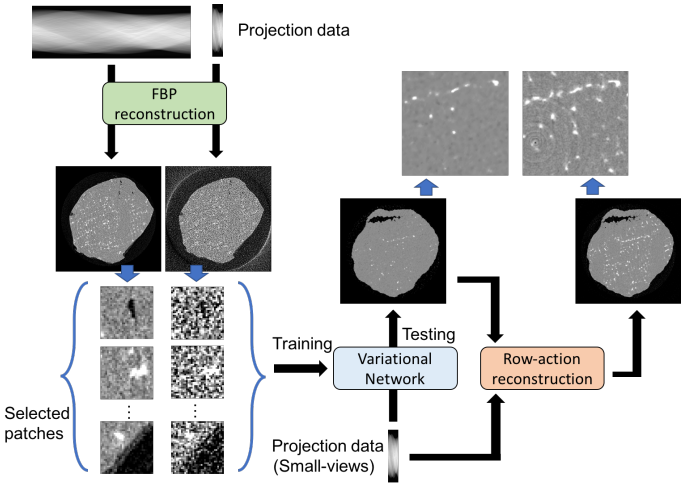

Fig. 1. A sketch demonstrate the proposed method. Two projection data of different sizes are used to produce images with different quality through FBP reconstruction. Image patches are used to train the VN. Testing small-views projection data is passed to VN to generate initial image, which is used with projection data for row-action reconstruction.

I. Introduction

Synchrotron radiation (SR) is a leading technology in micro-tomographic imaging. It can provide high-quality imaging of small objects with spatial resolution less than 1 μm. However, it is well-known that high-resolution SR imaging requires a relatively large number of projection views to avoid significant noise and artifacts.

Deep learning (DL) is an emerging machine learning technique that has recently attracted much attention [1]. It has been used extensively in solving image and signal processing problems and tomographic imaging [2]. Variational networks (VN) is presented recently as an approach for image restoration/denoising problems [3]. The development of VN is motivated by the proximal gradient and proximal incremental methods. The powerful of VN in noise suppression motivate us to extend it into a more practical image reconstruction algorithm for SR imaging. In image reconstruction, DL techniques are commonly used as a post-processing step applied directly to the reconstructed image, without consideration of the projection data used for the reconstruction [4], [5]. This may lead to the loss of important image information as the original projection data is used for reconstruction but not for restoration from noise and other defects. In this paper, a novel procedure for the use of VN in image reconstruction from limited projection data is proposed. The demonstration here is limited to the case of small-views problem. However, the proposed method is valid for other data limitation problems such as interior tomography and the limited-angle problem. without consideration of data fitting.

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II. Methods

The proposed method consists of two phases; training and testing. Patches extracted from images reconstructed from small/large number of views are used to train the VN. Image formation in testing phase is achieved through two steps. First, the trained VN is used to estimate the initial image which is highly dependent on the structure of the network, training dataset and several fine tuning parameters. Second, the initial image is subject to further improvement using a row-action image reconstruction algorithm that fits the initial estimation with the measured projection data (Fig. 1). The second step ensures the consistency between the reconstructed image and the acquired data. Thus, the final image is expected to avoid drawbacks of alternative DL techniques that is processed with image domain as a post-processing step similar to image denoising techniques.

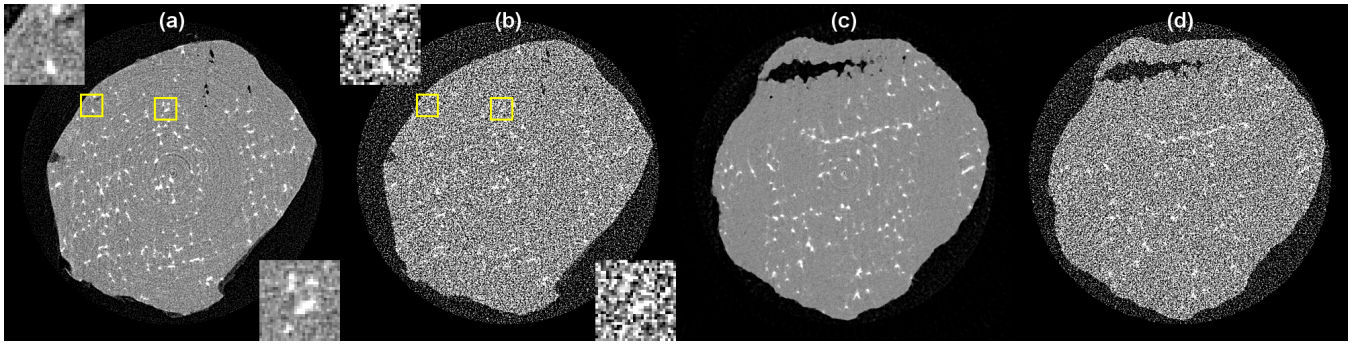

Fig. 2. Images representing a single slice reconstructed from (a) 1500 and (b) 100 projection views using FBP algorithm. The yellow square regions are magnified in the top-left and bottom-right corners of each image. A reconstruction of different object slice obtained using 100 projections in (c) with corresponding FBP image in (d).

A. Initial estimation with VN

Variational networks are developed to have a similar computational structure with the well-known residual networks. At the same time, VN is presented within the framework of variational methods that is proved to be very efficient in several image restoration/reconstruction applications such as Total Variation (TV) regularization. Inspired by the work in [3], the VN is extended here for a more robust reconstruction with limited training resources and/or limited data measurements. The image reconstruction problem problem is formulated as follows:

$$\min_{\vec{u}} Q(\vec{u}) = \sum_{c=1}^{C} q_c(\vec{u}, \vec{\theta}_c) = R_c(\vec{u}; \vec{\theta}_c) + D_c(\vec{u}; \vec{\theta}_c), \quad (1)$$

where $\vec{u} = (u_1, \ldots, u_n)^\top$ is an image vector, $\vec{\theta}$ is the parameter vector corresponding to each component c, R_c is a regularizer term enforce some prior knowledge and D_c is the data fidelity term. The solution of problem in Eq. (1) is found through an incremental proximal gradient method as detailed in [3] and resulting initial image estimate $\vec{u}$ is passed to the next step for further improvement.

B. Row-action reconstruction algorithm

The initial estimate $\vec{u}$ obtained from Eq. (1) is highly dependent on the training strategy that is used to estimate R_c. In case where a large enough training data set is not available, such as the case of SR imaging, we improve $\vec{u}$ through further reconstruction step that significantly improve the image quality. The estimated VN image $\vec{u}$ is expected to be close enough to the true image such that the difference is sparse. The reconstruction step is based on minimizing the following cost function:

$$\|\vec{x} - \vec{u}\|_p^p / p \text{ subject to } A\vec{x} = \vec{b},\ \vec{x} \geq 0, \quad (1 < p \leq 2). \quad (2)$$

where $\vec{b} = (b_1, \ldots, b_m)^\top$ is the measured projection data and $A = \{a_{ij}\}$ is $m \times n$ matrix that model the SR imaging system. As $m \ll n$, there are several feasible solutions of Eq. (2). The Lagrangian duality is used to solve Eq. (2). Given the following optimization problem with linear equality constraint:

$$\text{minimize } F(\vec{x}) \text{ subject to } A\vec{x} = \vec{b}, \quad (3)$$
$$F(\vec{x}) = \|\vec{x} - \vec{u}\|_p^p / p.$$

The associated Lagrangian dual problem is given by:

$$\text{maximize } D(\vec{\lambda}), \quad (4)$$
$$D(\vec{\lambda}) = \inf_{\vec{x} \in \mathbf{dom}(F)} \mathcal{L}(\vec{x}, \vec{u}, \vec{\lambda}),$$

where $\vec{\lambda} = (\lambda_1, \ldots, \lambda_m)^\top$ is the dual variables' vector and $\mathcal{L}(\vec{x}, \vec{u}, \vec{\lambda})$ is the Lagrangian function associated with Eq. (3), which defined as:

$$\mathcal{L}(\vec{x}, \vec{u}, \vec{\lambda}) = F(\vec{x}) - \lambda^\top (A\vec{x} - \vec{b}). \quad (5)$$

and the conjugate function of $F(\vec{x})$ is then defined as:

$$G(\vec{y}) = \sup_{\vec{x} \in \mathbf{dom}(F)} \left(\vec{y} \cdot \vec{x} - F(\vec{x}) \right). \quad (6)$$

Therefore, the reconstruction algorithm is implemented as follows. First, we iteratively solve the maximization problem in Eq. (4) using coordinate ascent method to obtain the dual solution $\vec{\lambda}^*$ of the maximization problem. Then, the following primal-dual relation $x_j^* = g'_{u_j}(A^\top \vec{\lambda}^*|_j)$ is used to find the corresponding solution of the original problem in Eq. (2). More details about the derivation of the row-action reconstruction algorithm can be found in our earlier publication in [6].

III. Experimental Results

Experimental data obtained from SPring-8 synchrotron radiation (Hyogo, Japan) was used to evaluate the proposed framework. The object is aluminum alloy (Al5Cu) of size 1.0 mm, which was imaged using beamline BL20XU. The object was scanned with x-ray energy of 35 keV over 1500 projection views with angular spacing of 0.12°. Projection data is measured using a CCD camera of 4000×2624 pixels with 2×2 binning mode and 0.5×0.5 μm pixel size. The projection

data was down-sampled into 100 projection views (1/15 of original data measurements) to simulate the case of small-views data acquisition. Image reconstructed from different projection samples using standard FBP algorithm are shown in Fig. 2. It is clearly seen that the image reconstructed from small views is suffering from severe noise that complicate the detection of the fine image details. A small set of image pixels are used to train the VN. The training dataset of 1800 pixels of a small-views image and their corresponding eight neighbor pixels in full-views image are used as input patterns for the training phase. A sample of those pixels are shown in the magnified regions in Fig. 2(a) and (b). The network is tested with an input using full image shown in Fig. 2(d) and final reconstruction is shown in Fig. 2(c) where a superior image quality is observed.

IV. Conclusion

This work presents a framework for image reconstruction from limited projection data with the assistance of VN. The power of the proposed method is recognized in the cases where the training data is relatively small such that implementing DL approach only to the reconstructed image is not enough to achieve high quality reconstruction. Also, it guarantees the consistency of reconstructed image and projection data which is likely preserve any potential abnormalities that might be diminished if only DL approach is used. The proposed method is evaluated using synchrotron radiation real data and high-quality images can be achieved with small number of projection views corresponding to only 1/15 of the originally required data for conventional reconstruction.

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